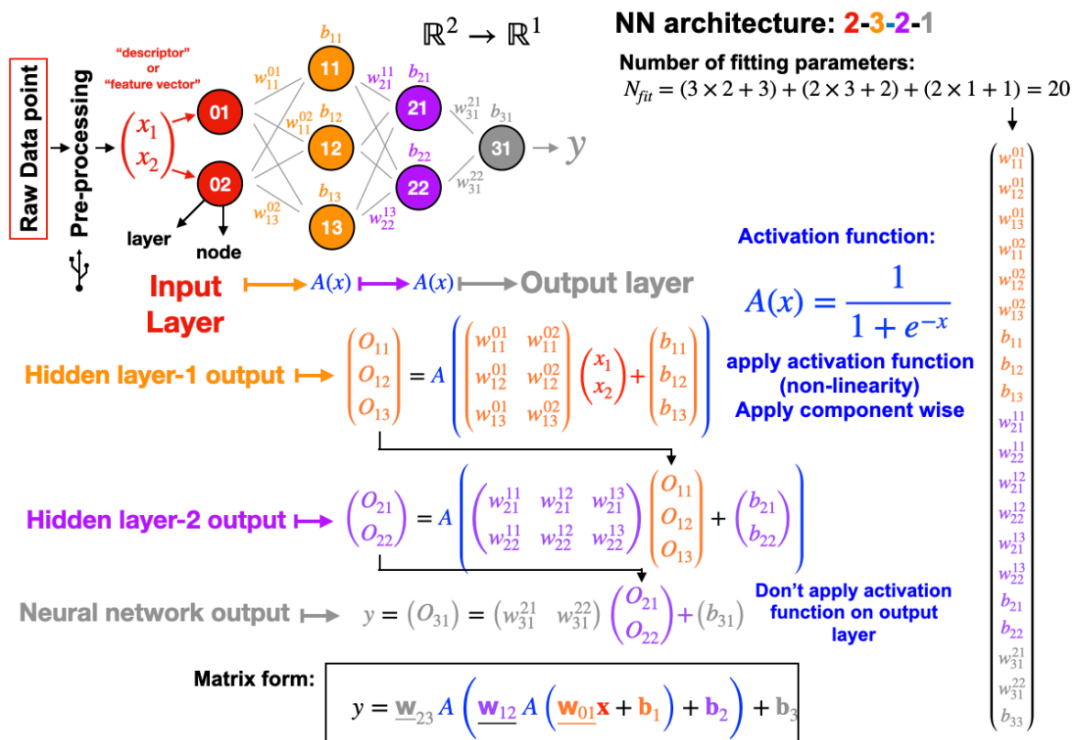


Study Guide: ANNs

1. **ANN evaluation:** Be able to numerically evaluate a small ANN model with provided numerical values for inputs and weight matrices. Similar to the following

- **Problem Type:** Scalar regression (unconstrained output)
- **Mapping:** $\mathbb{R}^2 \rightarrow \mathbb{R}^1$ (input layer) $\mathbb{R}^1$ (output layer)
- **Hidden layers:** $N_{hidden} = 2 \rightarrow [3,2]$
- **Hidden-layer activation:** Sigmoid
- **Output-layer activation:** Linear
- **Number of fitting parameters:** 20



2. Neural Network Design: Fill-in-the-Blank

Given a description of a dataset and a corresponding learning task. Focus on matching the network design choices to the data and type of problem (e.g., regression, binary classification, multi-class classification). Based on this information, determine the appropriate neural network configuration by filling in the following components:

- **Input layer size:** number of features provided to the model
- **Output layer size:** number of predicted values or classes
- **Output layer activation function:** e.g., sigmoid, softmax, linear
- **Loss function:** appropriate to the task (e.g., binary cross-entropy, categorical cross-entropy, mean squared error)
- For examples see: <https://jfh.georgetown.domains/centralized-lecture-content/content/machine-learning/deep-learning/fundamentals/introduction-to-neural-networks/slides.html#summary-of-ann-type-models>

3. **Fitting parameter calculation:** Given an ANN architecture be able to tell the number of fitting parameters of each of the various learning parameter matrices.
4. **Activation functions:** Be able to plot and/or describe use-cases for the various activation functions provided in the slides <https://jfh.georgetown.domains/centralized-lecture-content/content/machine-learning/deep-learning/fundamentals/introduction-to-neural-networks/slides.html#activation-functions>
5. **ANN training:** Understand the standard training workflow for parametric learning in neural networks, including the forward pass, loss computation, gradient-descent step, and parameter updates.
 - Interpret the training diagram (data → predictions → loss → gradients → parameter update)
 - Work with provided gradient matrices for a given weight matrix
 - Perform a single step of **gradient descent**, updating weights and/or biases using the learning rate and computed gradients

